

Department of Electronics and Communication Engineering

Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. ECE A&B

Course Code & Title: EC8651 Transmission Lines and RF Systems

Name of the Faculty member: Mr.D.Gopinath

Innovative Practice Description

- **Unit / Topic: Unit II / Impedance matching methods**
- **Course Outcome: CO3** The students will be able to analyze impedance matching by stubs using smith charts.
- **Topic Learning Outcome: TLO 10** The students will be able to use Smith charts to conclude results related to Impedance matching
- **Activity Chosen: Student Team Achievement Divisions (STAD)**
- **Justification:**

Student Team Achievement Divisions (STAD) is a most suited collaborative learning model for a Conceptual course. The different types of impedance matching methods and its design procedure can be studied and learned using Smith chart. The impedance matching techniques are Quarter Wave Transformer, Single stub matching and Double stub matching. STAD is a Collaborative Group activity. We can assign different matching problems to the student groups to learn the design procedures better.

- **Time Allotted for the Activity: 50 minutes**

Details of the Implementation:

Step1: Group Study Time (25 minutes)

- ❖ The time is given for each group to revise/study the impedance matching techniques and its design procedures.
- ❖ Each group should study the Quarter Wave Transformer, Design procedure and Applications
- ❖ Each group should study the Single stub matching, Design procedure and Applications
- ❖ Each group should study the Double stub matching, Design procedure and Applications
- ❖ The high caliber students take the responsibility of ensuring that every member is well-prepared in the content.

Step 2 – Final Evaluation Quiz – (15 minutes)

- ❖ At the end of the stipulated time, the entire class taken a quiz. The quiz is corrected in class by projecting the correct answers.
- ❖ Individual students' scores have displayed and then group scores can be calculated.

Step 3 - Announcing the Winners (5 minutes)

- ❖ Top scoring group and Top performer has appreciated and encouraged by the Senior faculty member of the department.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PSO3
CO3	3	2	2	2	1	2	3	3

- **PO / PSO mapped:**

Innovative practice	PO1	PO8	PO9	PO10	PSO3
	3	3	3	3	2
Justification for correlation	Circuit theory fundamentals are applied	Group activity which requires inter-personal ethics	Needs individual and team work	Effective communication is required	Students involves design and analysis of different impedance matching techniques

- **Images / Screenshot of the practice:**





• **Reflective Critique:**

Benefits:

- ❖ The students involved activity in the activity and discussed with their neighbors.
- ❖ The understanding level of the concept can be identified, while the students are sharing their view.
- ❖ The question was asked in the Assessment test, most of the students answered correctly.

Challenges:

- ❖ Making all the students to share the view was not possible due to time constrain.
- ❖ While randomly choosing the group few members struggle to answer the question.

References:

1. <https://www.scirp.org/journal/paperinformation.aspx?paperid=96719>

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Department of Electronics and Communication Engineering

Academic Year 2022 – 2023 (Even Semester)

Active Learning Activity- Unit-I- Transmission Line Theory

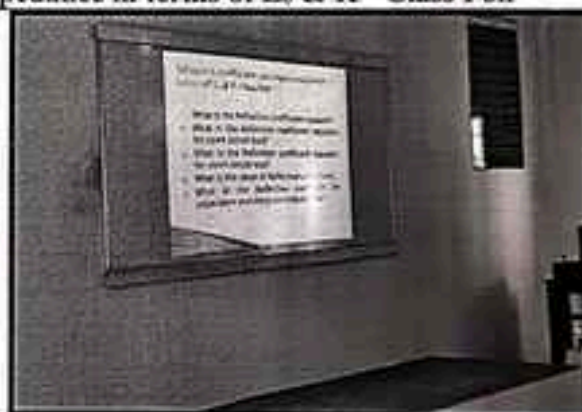
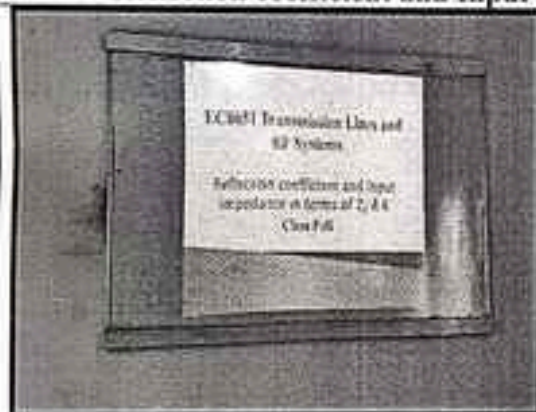
Degree, Semester & Branch: VI Semester B.E. ECE A

Course Code & Title: EC8651 Transmission Lines and RF Systems

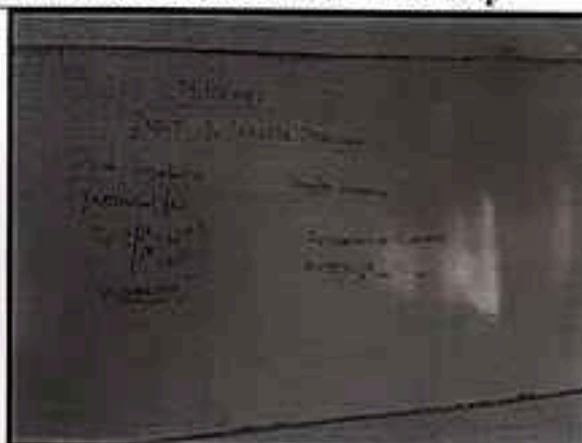
Name of the Faculty member: Mr.D.Gopinath

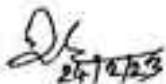
Name of the Topic	Name of the Activity	Date & Duration
Reflection coefficient and Input impedance in terms of Z_0 & K	Class Poll	10.02.2023 & 5 Minutes
Input and transfer impedance, Open and short circuited lines	Mind map	22.02.2023 & 5 Minutes

Reflection coefficient and Input impedance in terms of Z_0 & K - Class Poll



Input and transfer impedance, Open and short circuited lines - Mind map




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Active Learning Activity- Unit-II- High Frequency Transmission Lines

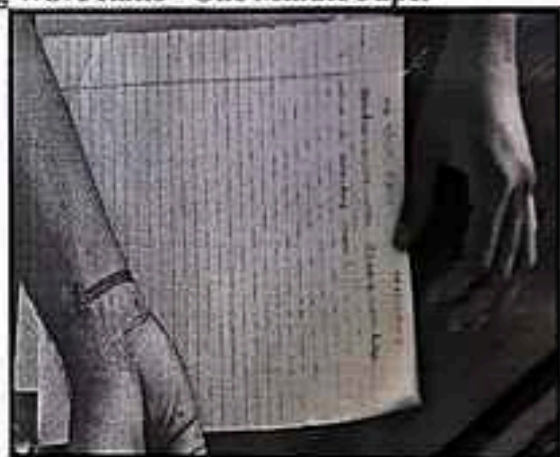
Degree, Semester & Branch: VI Semester B.E. ECE A

Course Code & Title: EC8651 Transmission Lines and RF Systems

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Standing Waves, Nodes, Standing Wave Ratio	One Minute Paper	28.02.2023 & 5 Minutes
Measurement of VSWR and wavelength, Revision	Exit Slip	09.03.2023 & 5 Minutes

Standing Waves, Nodes, Standing Wave Ratio - One Minute Paper



Measurement of VSWR and wavelength- Exit Slip



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Academic Year 2022 – 2023 (Even Semester)

Active Learning Activity- UNIT III IMPEDANCE MATCHING IN HIGH FREQUENCY LINES

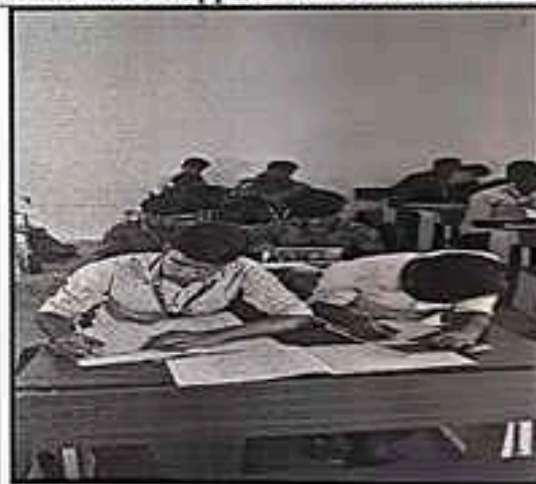
Degree, Semester & Branch: VI Semester B.E. ECE B

Course Code & Title: EC8651 Transmission Lines and RF Systems

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Impedance matching: Quarter wave transformer and its applications	Brain Storming	09.03.2023 & 5 Minutes
Single stub matching using Smith chart	Student Team Achievement Divisions (STAD)	27.03.2023 & 50 minutes

Impedance matching: Quarter wave transformer and its applications - Brain Storming




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Active Learning Activity- Unit-I- Transmission Line Theory

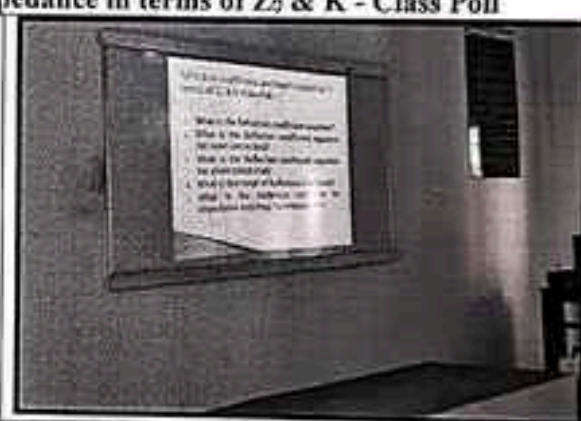
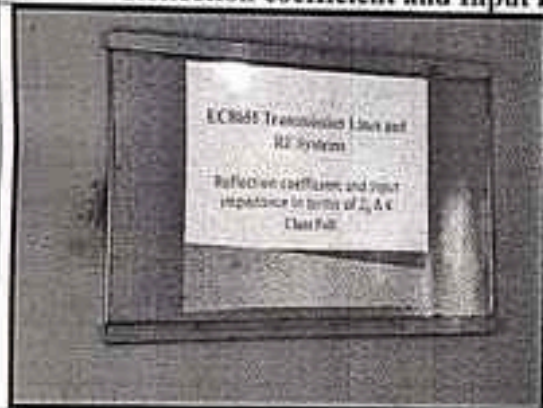
Degree, Semester & Branch: VI Semester B.E. ECE B

Course Code & Title: EC8651 Transmission Lines and RF Systems

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Reflection coefficient and Input impedance in terms of Z_0 & K	Class Poll	10.02.2023 & 5 Minutes
Input and transfer impedance, Open and short circuited lines	Mind map	22.02.2023 & 5 Minutes

Reflection coefficient and Input impedance in terms of Z_0 & K - Class Poll



Input and transfer impedance, Open and short circuited lines - Mind map




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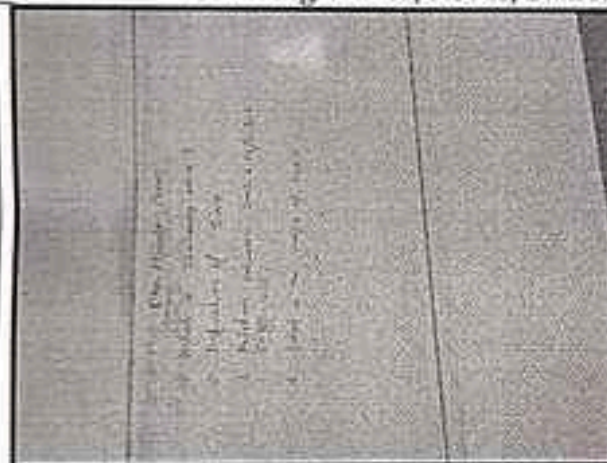
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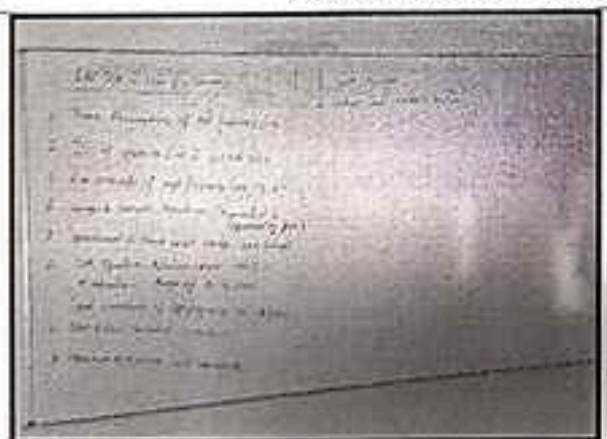
Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Standing Waves, Nodes, Standing Wave Ratio	One Minute Paper	27.03.2023 & 5 Minutes
Measurement of VSWR and wavelength, Revision	Exit Slip	08.03.2023 & 5 Minutes

Standing Waves, Nodes, Standing Wave Ratio - One Minute Paper



Measurement of VSWR and wavelength- Exit Slip



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Active Learning Activity- Unit-III- Impedance Matching in High Frequency Lines

Degree, Semester & Branch: VI Semester B.E. ECE A

Course Code & Title: EC8651 Transmission Lines and RF Systems

Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Impedance matching: Quarter wave transformer and its applications	Brain Storming	13.03.2022 & 5 Minutes
Single stub matching using Smith chart	Student Team Achievement Divisions (STAD)	28.03.2023 & 50 minutes

Impedance matching: Quarter wave transformer and its applications - Brain Storming



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